Rank of the Symmetrizing Product: Proof, Correction, and Computation

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Abstract

We study the rank of the element $\Pi = \prod_{j=1}^{m}(1 + s_{i_j}) \in \mathbb{C}[\mathfrak{S}_n]$ where $s_{i_1} \cdots s_{i_m}$ is a reduced word for the longest element $w_0 \in \mathfrak{S}_n$. We prove that for the Iwahori–Hecke algebra $H_q(\mathfrak{S}_n)$ at generic q, the rank of the analogous product is q-independent (Theorem 4). We verify computationally for $n \leq 8$ that the staircase reduced word gives total rank $n!/2^{\lfloor n/2 \rfloor}$ and establish a hook formula for the rank in hook representations (Theorem 8, verified $n \leq 9$). Critically, we **correct** a prior claim: the rank is **not** word-independent. Different reduced words for w_0 can yield different rank vectors, and this is already visible at $n = 4$.

1 Setup and notation

Let $\mathfrak{S}_n$ denote the symmetric group on $\{1, \ldots, n\}$ with simple transpositions $s_i = (i, i{+}1)$ for $i = 1, \ldots, n{-}1$. The longest element $w_0 = [n, n{-}1, \ldots, 1]$ has length $\ell(w_0) = \binom{n}{2}$.

Definition 1 (Staircase word)**.** The *staircase reduced word* for $w_0 \in \mathfrak{S}_n$ is

$$\mathbf{i}_{\text{stair}} = (s_1)(s_2, s_1)(s_3, s_2, s_1) \cdots (s_{n-1}, s_{n-2}, \ldots, s_1).$$

Explicitly, batch k consists of the generators $s_k, s_{k-1}, \ldots, s_1$ for $k = 1, \ldots, n{-}1$. The total length is $\binom{n}{2}$.

Definition 2 (Symmetrizing product)**.** For a reduced word $(i_1, \ldots, i_m)$ for w_0, define

$$\Pi = \prod_{j=1}^{m}(1 + s_{i_j}) \in \mathbb{C}[\mathfrak{S}_n].$$

Equivalently, $\Pi = 2^m \prod_{j=1}^{m} P_{i_j}$ where $P_i = (1 + s_i)/2$ is the orthogonal projection onto the $(+1)$-eigenspace of s_i.

The rank of Π acting on an irreducible representation V_λ is denoted r_λ, and the total rank in the left regular representation is $\sum_\lambda r_\lambda \cdot f^\lambda$ where $f^\lambda = \dim V_\lambda$.

2 Correction: Word dependence of rank

The rank vector $(r_\lambda)_\lambda$ depends on the choice of reduced word.

This was claimed to be word-independent in a prior analysis, based on checking only the staircase word and the reverse staircase word (which happen to agree). Exhaustive computation over all reduced words reveals:

- $\mathfrak{S}_4$: There are 16 reduced words for w_0. Of these, 12 give total rank $6 = 4!/4$, and 4 give total rank 12. The 4 "anomalous" words are precisely those where s_2 appears 3 times (e.g., $(1, 2, 3, 2, 1, 2)$). In the standard representation $V_{(3,1)}$, the good words give rank 1 while the anomalous words give rank 2.
- $\mathfrak{S}_5$: There are 768 reduced words. Of these, 764 give total rank $30 = 5!/4$, and 4 give total rank 60.

The staircase word and the reverse staircase word are always "good" (giving the minimal rank). All results below refer to the **staircase word** unless otherwise stated.

Remark 3. The anomalous words seem to be those where some "middle" generator is over-represented relative to the staircase frequency. Understanding exactly which reduced words give the minimal rank is an open question.

3 q-Independence (Proved)

Theorem 4 (q-independence of rank)**.** *Let $H_q(\mathfrak{S}_n)$ be the Iwahori–Hecke algebra with generators $T_1, \ldots, T_{n-1}$ satisfying $(T_i - q)(T_i + 1) = 0$ and the braid relations. For any reduced word $(i_1, \ldots, i_m)$ for w_0, define*

$$\Pi_q = \prod_{j=1}^{m} (T_{i_j} + 1).$$

At generic q (avoiding 0, -1, and roots of unity), the rank of Π_q in each irreducible representation of $H_q(\mathfrak{S}_n)$ equals the rank of the corresponding product $\Pi_1 = \prod(1 + s_{i_j})$ in $\mathbb{C}[\mathfrak{S}_n]$.

Proof. Define $P_i^{(q)} = (T_i + 1)/(q + 1)$. Then

$$(P_i^{(q)})^2 = \frac{(T_i + 1)^2}{(q + 1)^2} = \frac{T_i^2 + 2T_i + 1}{(q + 1)^2} = \frac{(q - 1)T_i + q + 2T_i + 1}{(q + 1)^2} = \frac{(q + 1)T_i + (q + 1)}{(q + 1)^2} = \frac{T_i + 1}{q + 1} = P_i^{(q)},$$

so each $P_i^{(q)}$ is idempotent.

By the Tits deformation theorem, for generic q there exists an algebra isomorphism $\varphi : H_q(\mathfrak{S}_n) \xrightarrow{\sim} \mathbb{C}[\mathfrak{S}_n]$ that preserves the block structure (sending irreducibles to irreducibles of the same dimension). Since $P_i^{(q)}$ is the unique idempotent in H_q projecting onto the trivial representation of $\langle T_i \rangle \cong H_q(\mathfrak{S}_2)$, and $P_i^{(1)} = (1 + s_i)/2$ is the corresponding idempotent in $\mathbb{C}[\mathfrak{S}_n]$, the isomorphism sends $P_i^{(q)} \mapsto P_i^{(1)}$.

Since $\Pi_q = (q + 1)^m \prod P_{i_j}^{(q)}$, the isomorphism sends $\Pi_q/(q + 1)^m$ to $\prod P_{i_j}^{(1)} = \Pi_1/2^m$. Algebra isomorphisms preserve matrix rank in each simple component. □

4 Rank formula for the staircase word

Conjecture 5 (Total rank)**.** *For the staircase reduced word, the total rank of Π in the left regular representation of $\mathfrak{S}_n$ is*

$$\operatorname{rank}(\Pi) = \frac{n!}{2^{\lfloor n/2 \rfloor}}.$$

Verification. Computed directly for $n = 2, 3, \ldots, 8$:

n	2	3	4	5	6	7	8
$n!/2^{\lfloor n/2 \rfloor}$	1	3	6	30	90	630	2520
Computed rank	1	3	6	30	90	630	2520

The ratio $\operatorname{rank}(n)/\operatorname{rank}(n-1)$ is n when n is odd and $n/2$ when n is even.

5 Vanishing criterion

Conjecture 6 (Vanishing)**.** *For the staircase word, $r_\lambda = 0$ if and only if $\ell(\lambda) > (n+1)/2$, where $\ell(\lambda)$ is the number of parts of λ.*

Verification. Confirmed for all partitions of $n \leq 8$.

Remark 7. Self-conjugate partitions always survive: $\ell(\lambda) = \lambda_1$ and $n \geq \ell^2$, so $\ell \leq \sqrt{n} < (n+2)/2$ for $n \geq 3$. The sign representation $\lambda = (1^n)$ with $\ell = n$ is always killed (each factor $(1 + s_i)$ annihilates it since s_i acts as -1). For the anomalous words of Section 2, the vanishing criterion fails: previously-killed representations can acquire nonzero rank.

6 Hook formula (verified)

Theorem 8 (Hook representation rank)**.** *For the staircase reduced word and hook partitions $\lambda = (n-k, 1^k)$ with $0 \leq k \leq n-1$,*

$$r_{(n-k,1^k)} = \binom{\lfloor (n-1)/2 \rfloor}{k}.$$

In particular, $r_{(n-k,1^k)} = 0$ when $k > \lfloor (n-1)/2 \rfloor$, consistent with the vanishing criterion.

Verification. Computed for all hooks of $\mathfrak{S}_n$, $2 \leq n \leq 9$: all 42 values match exactly.

Remark 9. The standard representation $V_{(n-1,1)}$ has rank $\lfloor (n-1)/2 \rfloor$. In the permutation representation $\mathbb{C}^n = V_{(n)} \oplus V_{(n-1,1)}$, the total rank is $1 + \lfloor (n-1)/2 \rfloor = \lceil (n+1)/2 \rceil$, which counts the dimension of the "partially sorted" subspace.

Remark 10. Although $V_{(n-k,1^k)} \cong \bigwedge^k V_{(n-1,1)}$ as $\mathfrak{S}_n$-representations, the formula $r_{(n-k,1^k)} = \binom{r_{(n-1,1)}}{k}$ does **not** follow formally from exterior algebra. The action of the group algebra element Π on $\bigwedge^k V$ is $\sum_w c_w \bigwedge^k \rho(w)$, which differs from $\bigwedge^k (\sum_w c_w \rho(w))$. The hook formula is a *genuine* property of the staircase product, not a formal consequence.

7 Full irrep decomposition

The rank r_λ for **all** non-killed irreps of $\mathfrak{S}_n$ ($n \leq 8$, staircase word):

$\mathfrak{S}_5$: $(5)\to 1$, $(4,1)\to 2$, $(3,2)\to 2$, $(3,1^2)\to 1$, $(2^2,1)\to 1$.
$\mathfrak{S}_6$: $(6)\to 1$, $(5,1)\to 2$, $(4,2)\to 3$, $(4,1^2)\to 1$, $(3^2)\to 1$, $(3,2,1)\to 2$, $(2^3)\to 1$.
$\mathfrak{S}_7$: $(7)\to 1$, $(6,1)\to 3$, $(5,2)\to 5$, $(5,1^2)\to 3$, $(4,3)\to 4$, $(4,2,1)\to 6$, $(4,1^3)\to 1$, $(3^2,1)\to 3$, $(3,2^2)\to 3$, $(3,2,1^2)\to 2$, $(2^3,1)\to 1$.
$\mathfrak{S}_8$: $(8)\to 1$, $(7,1)\to 3$, $(6,2)\to 6$, $(6,1^2)\to 3$, $(5,3)\to 6$, $(5,2,1)\to 8$, $(5,1^3)\to 1$, $(4^2)\to 3$, $(4,3,1)\to 7$, $(4,2^2)\to 6$, $(4,2,1^2)\to 3$, $(3^2,2)\to 3$, $(3^2,1^2)\to 2$, $(3,2^2,1)\to 3$, $(2^4)\to 1$.

A closed-form formula for general r_λ remains open. Among non-killed partitions, the "boundary" cases $\lambda = (2^{\lfloor n/2 \rfloor})$ (for n even) and $\lambda = (2^{\lfloor n/2 \rfloor}, 1)$ (for n odd) always have $r_\lambda = 1$.

8 Proof sketch for the standard representation rank

We prove $r_{(n-1,1)} = \lfloor (n-1)/2 \rfloor$ for the staircase word.

Proof sketch. Work in the standard representation $V = \{x \in \mathbb{R}^n : \sum x_i = 0\}$, dimension $n-1$. The projection P_i maps $(x_1, \dots, x_n) \mapsto (\dots, \frac{x_i+x_{i+1}}{2}, \frac{x_i+x_{i+1}}{2}, \dots)$, projecting onto the hyperplane $\{x_i = x_{i+1}\}$.

The staircase product applies batches: batch k is $P_k P_{k-1} \cdots P_1$. Each batch ends with P_1, so the image after every batch satisfies $x_1 = x_2$.

Dimension tracking. Starting dimension is $n-1$. At batch k, the leading projection P_k is applied to the current image W. The dimension drops by $\dim(\ker P_k \cap W)$, which is 0 or 1.

Empirically (verified for $n \leq 9$): the dimension drops by exactly 1 at odd batches ($k = 1, 3, 5, \dots$) and stays constant at even batches ($k = 2, 4, 6, \dots$). The number of odd batches is $\lceil (n-1)/2 \rceil = \lfloor n/2 \rfloor$. The final dimension is $(n-1) - \lfloor n/2 \rfloor = \lfloor (n-1)/2 \rfloor$.

A complete proof of this dimension-drop pattern requires showing that the image after batch $k-1$ has a 1-dimensional intersection with $\ker P_k$ when k is odd, and trivial intersection when k is even. This remains an open step. □

9 Gaps and open questions

1. **Total rank formula.** We conjecture $\mathrm{rank}(\Pi_{\mathrm{stair}}) = n!/2^{\lfloor n/2 \rfloor}$ but lack a proof. The inductive structure (ratio n or $n/2$) suggests an approach via the branching rule $\mathfrak{S}_n \downarrow \mathfrak{S}_{n-1}$.

2. **Characterizing good words.** Which reduced words for w_0 give the minimal rank $n!/2^{\lfloor n/2 \rfloor}$? The staircase and reverse staircase always do. For $\mathfrak{S}_4$, exactly 12/16 words are good; for $\mathfrak{S}_5$, 764/768.

3. **General r_λ formula.** The hook formula $r_{(n-k,1^k)} = \binom{\lfloor (n-1)/2 \rfloor}{k}$ is clean, but no formula is known for general λ. The two-row sequence $r_{(n-2,2)} = 1, 2, 3, 5, 6$ for $n = 4, \dots, 8$ does not match any standard combinatorial sequence.

4. **Standard representation proof.** The dimension-drop pattern at odd batches is verified numerically but requires a geometric/algebraic proof of why $\ker P_k \cap W_{k-1}$ alternates between dimension 0 and 1.

5. **Vanishing criterion.** The condition $\ell(\lambda) > (n+1)/2$ is verified for $n \leq 8$ but not proved. An inductive proof via branching fails at even n because the borderline case $\ell(\mu) = n/2$ escapes the inductive hypothesis.

10 Connection to Galashin's Yang–Baxter basis

The product $\Pi_{\mathrm{stair}}/2^m = \prod P_{i_j}$ equals $Y^{w_0}(p{=}1/2)$ in Galashin's Yang–Baxter basis notation [1]. The rank $n!/2^{\lfloor n/2 \rfloor}$ is thus a new invariant of this basis element. Galashin considers positivity and Schur expansion properties but does not study rank.

The word dependence discovered here has implications for the Yang–Baxter basis: the YB element at $p = 1/2$ is word-specific, and its rank encodes information about the reduced word itself.

References

[1] P. Galashin, *Symmetries of stochastic colored vertex models*, Ann. Probab. **49** (2021), no. 5, 2175–2219. arXiv:2003.06330.